
Project: How does the performance of a Greenpower car change during a race?

I am going to use fake data to analyse how the performance of a Greenpower car change during a race.

Main objectives:

- See what conclusions can be made about the efficiency of empres.
- Identify the source of what causes empres to slow down during race.
- Use the data to make the process of looking for solutions that allow empres to maintain speed for as long as possible during races easier.

Display Table:

```
GreenpowerRaceData_fake_ = readtable('Greenpower race data (fake).xlsx');  
GreenpowerRaceData_fake_
```

*Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.*

```
GreenpowerRaceData_fake_ =
```

15×7 table

	LapNumber	LapTime_secs_	Var3	BatteryVoltage_V_
	BatteryCurrent_A_	TrackDistance_metres_		TotalRaceTime_mins_
28	1	220.4	NaN	24
		3200		120
NaN	2	239.4	NaN	NaN
		NaN		NaN
NaN	3	260.8	NaN	NaN
		NaN		NaN
NaN	4	279.7	NaN	NaN
		NaN		NaN
NaN	5	300.5	NaN	NaN
		NaN		NaN
NaN	6	319.3	NaN	NaN
		NaN		NaN
NaN	7	340.2	NaN	NaN
		NaN		NaN
	8	360.9	NaN	NaN

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NaN		NaN		NaN
	9	379.6	NaN	NaN
NaN		NaN		NaN
	10	399.2	NaN	NaN
NaN		NaN		NaN
	11	420.3	NaN	NaN
NaN		NaN		NaN
	12	440.6	NaN	NaN
NaN		NaN		NaN
	13	459.5	NaN	NaN
NaN		NaN		NaN
	14	480.1	NaN	NaN
NaN		NaN		NaN
	15	499.5	NaN	NaN
NaN		NaN		NaN

Import data from table into variables:

```
lapNumber = GreenpowerRaceData_fake_ .LapNumber
lapTime = GreenpowerRaceData_fake_ .LapTime_secs_
batteryVoltage = GreenpowerRaceData_fake_ .BatteryVoltage_V_
batteryCurrent = GreenpowerRaceData_fake_ .BatteryCurrent_A_
lapDistance = GreenpowerRaceData_fake_ .TrackDistance_metres_
totalRaceTime = GreenpowerRaceData_fake_ .TotalRaceTime_mins_
```

lapNumber =

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

lapTime =

220.4000
239.4000
260.8000
279.7000
300.5000
319.3000
340.2000

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360.9000
379.6000
399.2000
420.3000
440.6000
459.5000
480.1000
499.5000

batteryVoltage =

24
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN

batteryCurrent =

28
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN

lapDistance =

3200
NaN
NaN
NaN

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NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN

totalRaceTime =

120
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN
NaN

Calculate Average lap time:

avgLapTime = mean(*lapTime*)

avgLapTime =

360

Fastest lap:

fastestLap = min(*lapTime*)

fastestLap =

220.4000

Slowest lap:

slowestLap = max(*lapTime*)

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```
slowestLap =
```

```
499.5000
```

Difference between fastest and slowest lap:

```
diff = slowestLap - fastestLap
```

```
diff =
```

```
279.1000
```

Calculate the speed of each lap:

```
lapDistanceDivider =
```

```
[3200;3200;3200;3200;3200;3200;3200;3200;3200;3200;3200;3200;3200;3200;3200]
```

```
eachLapSpeed = lapDistanceDivider./lapTime
```

```
lapDistanceDivider =
```

```
3200
```

```
3200
```

```
3200
```

```
3200
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3200
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3200
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3200
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3200
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3200
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3200
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3200
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```
3200
```

```
3200
```

```
eachLapSpeed =
```

```
14.5191
```

```
13.3668
```

```
12.2699
```

```
11.4408
```

```
10.6489
```

```
10.0219
```

```
9.4062
```

```
8.8667
```

```
8.4299
```

```
8.0160
```

```
7.6136
```

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7.2628
6.9641
6.6653
6.4064

Calculate Average lap speed:

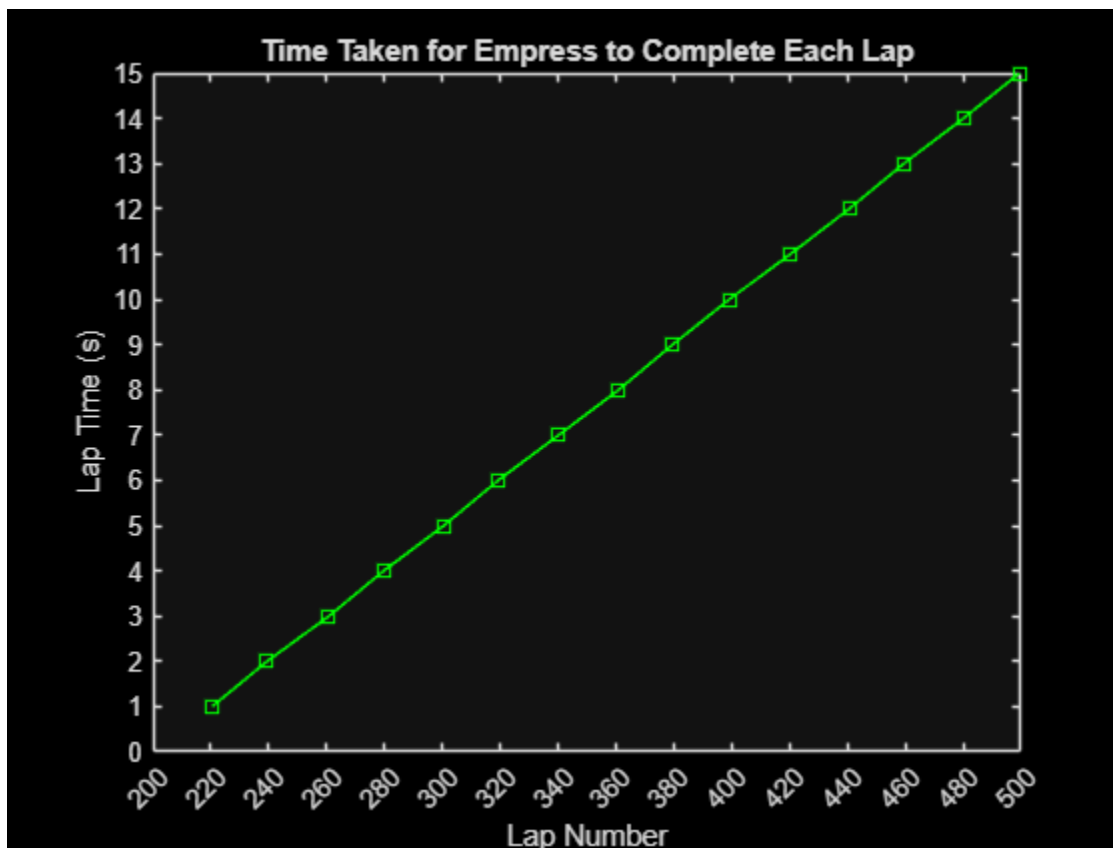
`avgLapSpeed = lapDistance(1)/avgLapTime`

avgLapSpeed =

8.8889

Graph to display the time taken for empress to complete each lap:

```
plot(lapTime,lapNumber,"gs-", LineWidth = 1)
xlabel("Lap Number")
ylabel("Lap Time (s)")
title("Time Taken for Empress to Complete Each Lap")
set(gca, "ytick", 0:1:15)
set(gca, "xtick", 0:20:500)
```



Graph to display the speed empress had each lap:

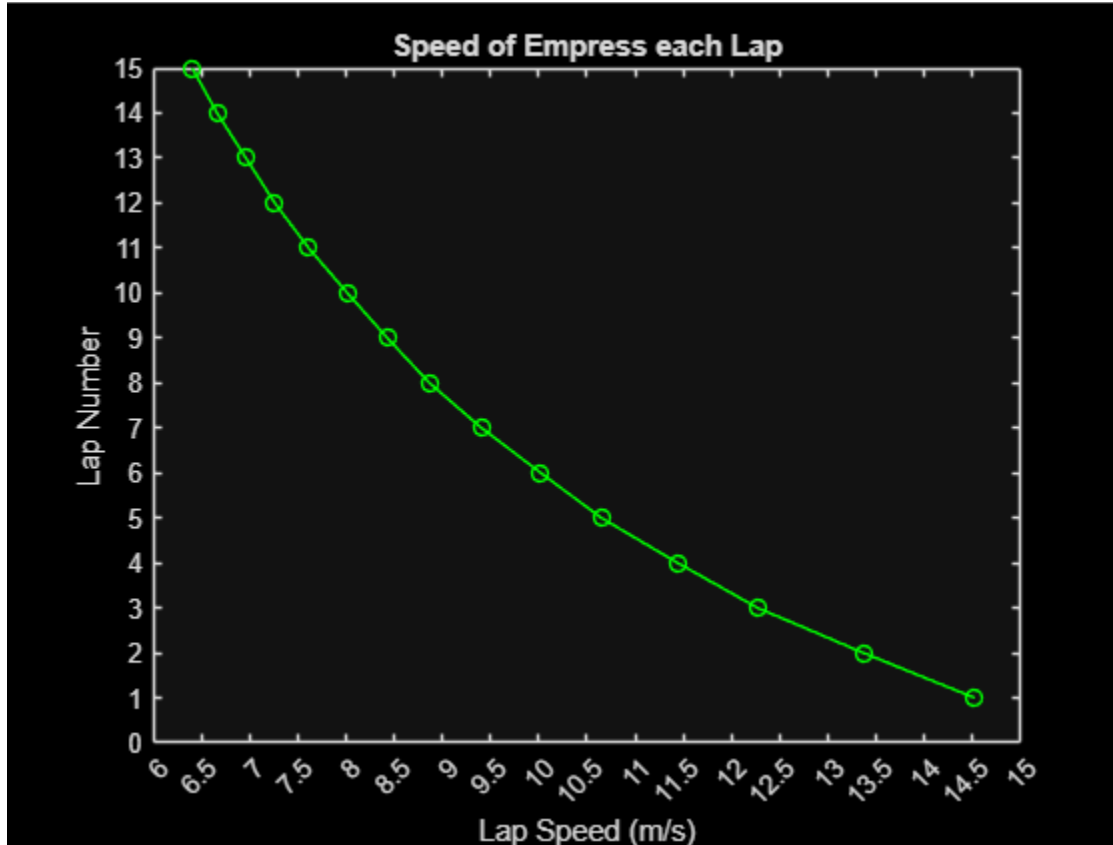
```
plot(eachLapSpeed,lapNumber,"go-", LineWidth = 1)
xlabel("Lap Speed (m/s)")
```

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```
ylabel("Lap Number")
title("Speed of Empress each Lap")
set(gca, "xtick", 0:0.5:15)
set(gca, "ytick", 0:1:15)

% *Analysis of graphs:*
% *Did performance improve or decline?*
%
% Empress' performance declined steadily
%
% *Were the lap times consistent?*
%
% The strong positive correlation between lap time and lap number show the lap
% times were not consistent. Empress got slower and slower as the race went
% on.
%
% *At what point did performance change?*
%
% The speed of empress decreased quite steadily from the first lap. However,
% this is probably because the data is fake. In real races I have been to
% where
% empress fails to maintain initial speed, usually the time taken for the car
% to complete each lap is around the same for the first 30 minutes then
% gradually
% gets slower and slower until it's walking pace.
%
% *What might have caused the change?*
```

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- The battery started at low current
- A driver could have braked often which drains the battery

Possible solutions?

- Use a battery with higher current e.g. 32A
- Encourage drivers not to brake as much as possible

• ? Final conclusions from this project:Benefits:

While it took longer to make calculations and draw graphs from data using MATLAB rather than excel it allowed me to develop the coding skills I learnt from the MATLAB course I did in a practical and fun way. There are lots of parallels between coding in MATLAB and python, so it also helped me practice concepts in both coding languages. This is why I think it would be helpful for other students at Greenpower to explore.

Furthermore, MATLAB is a platform that is used to allow engineers and data analysts to make calculations and visualise data, which are two career paths I am highly interested in and now know I would have fun having either one of these careers because of the MATLAB course and this project.

In terms of answering my initial question (How does the performance of a Greenpower car change during a race?), MATLAB certainly helped me to do this. The graphs show a clear answer. More team collaboration will be essential to use this data to look for solutions that allow empress to maintain speed for as long as possible.

Next steps to improve this project:

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- Would like to a third graph showing power consumption over a race but would appreciate guidance about how to do this because I'm not sure if it's possible to record the current, voltage, resistance etc. of the batteries over a race.
- Look for more efficient ways to code. This should be achieved through reading through more of the MATLAB help centre documentation and practice.
- Collaborate with the team to get suggestions about how to improve empress' performance.

Export code and results as a pdf

```
export("greenpower_race_analyser.mlx")
```

ans =

```
'/MATLAB Drive/greenpower_race_analyser.pdf'
```

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